BMC Psychology

SYSTEMATIC REVIEW Open Access

Neurological and biological correlations of ODD with ADHD in children and adolescents: a systematic review

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Abstract

Background Attention Deficit Hyperactivity Disorder (ADHD) often coexists with Oppositional Defiant Disorder (ODD), exacerbating impairment. The distinct neurological and biological patterns that differentiate this comorbidity from ADHD alone remain unclear.

Method We conducted systematic review of multi model correlates differentiating ODD+ADHD from ADHD alone following PRISMA 2020 with databases like MEDLINE, EMBASE, CINAHL, PsycINFO and Web of Sciences until August 2025. 144 full texts were reviewed and 26 studies were selected for final inclusion after 1457 records with 37 other sources. Designs included cross-sectional case-control studies, longitudinal studies, and imaging based on the ABCD framework, while measures covered structural and functional MRI, diffusion MRI, resting and task fNIRS, EEG and ERP, HPA-axis and immune-metabolic biomarkers, and executive and cognitive indices. The evaluation of bias employed RoB 2, ROBINS-I, and JBI, while synthesis utilised Braun-Clarke thematic analysis.

Results Oppositionality involved limbic–striatal and cerebello-cortical differences, with ODD+ADHD showing greater executive and emotion-processing deficits than ADHD alone. ODD was linked to lower cortisol and reduced sympathetic reactivity, while ADHD showed higher cortisol and tryptophan–kynurenine shifts, with cytokines decreasing after methylphenidate.

Conclusion ODD along with ADHD showed a distinct neurobiological and physiological pattern from ADHD alone. Such pattern is marked by greater executive and emotional regulation deficits with unique stress-response patterns, highlighting the need for tailored assessments and intervention approaches.

Keywords ADHD, ODD, Neurobiology, Executive function, Cortisol, Connectivity, Cerebellum, Biomarkers

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Introduction

Attention Deficit Hyperactivity Disorder (ADHD) frequently coexists with Oppositional Defiant Disorder (ODD) in children and adolescents. Approximately 60% of individuals with ADHD were concurrently diagnosed with ODD [1]. The Diagnostic and Statistical Manual of Mental Disorders (DSM-5) delineates ADHD as marked by inattention and/or hyperactivity/impulsivity, while ODD is characterized by irritable and angry dispositions alongside defiant and disobedient behaviors. The presence of comorbid ODD in individuals with ADHD is associated with increased functional impairment and a poorer long-term prognosis [2–5]. Contemporary nosology defines ODD by a persistent angry or irritable mood, argumentative or defiant behavior, and vindictiveness lasting for a minimum of six months. This phenotype overlaps with, yet is not limited to, conduct disorder and often complicates ADHD presentations [6, 7].

It has been widely reported that ODD is associated with increased emotional lability and delinquency in children with ADHD [4, 8]. Cognitive inhibition and visuospatial episodic memory are associated with ADHD-symptoms, whereas cognitive flexibility is related to ODD-symptoms [9]. Neuroimaging studies demonstrate structural and functional brain changes in ADHD and ODD [10, 11]. ADHD with ODD reveals cortical thickening [12], cerebellum [13], and volume decreases in multiple brain areas [14]. Lower FA in left lateral frontotemporal white matter is associated with antisocial behavior [15], although the results are inconsistent [16], suggesting neural correlates of irritable and aggressive behaviors.

The accumulation of multimodal evidence suggests dysfunctions in fronto-striatal and fronto-limbic circuits across the spectra of disruptive behavior and ADHD. However, it remains uncertain whether children with ODD+ADHD exhibit additive, synergistic, or distinct neural signatures compared to "pure" ADHD or ODD/CD groups—an essential gap this review aims to address [17–19]. Large-scale consortia analyses yield substantial findings: an ENIGMA-ASB coordinated analysis encompassing 15 cohorts revealed significant reductions in cortical surface area and alterations in subcortical structures among youth with conduct disorder, highlighting effects relevant to externalizing liability in the context of ODD comorbidity [17]. In contrast, ENIGMA-ADHD identified diminished volumes in the amygdala, accumbens, and hippocampus, indicative of delayed subcortical maturation unaccounted for by medication [18]. Work at the network level adds to these regional findings: Dysconnectivity in the default mode network (DMN) and interference with task-positive systems are consistently observed in ADHD, likely contributing to executive control failures and social dysfunction, which may be exacerbated in individuals with both ODD and ADHD [20–22].

Meta-analytic task-fMRI evidence demonstrates that emotion-regulation demands recruit amygdala–prefrontal coupling and that irritability, a central feature of ODD, is associated with disrupted amygdala connectivity with regulatory prefrontal regions implicating a "hot EF" mechanism that is likely to exacerbate ADHD-related "cool EF" deficits [23–25]. Based on meta-analyses of inhibitory control, the deficiency in ODD/CD was partially ADHD-related and thus emphasizing comorbidity-stratified designs [19]. Stress-immune markers suggest decreased baseline levels and morning cortisol and abnormalities in cytokines among pediatric ADHD, being influenced by externalizing characteristics and adversity availabilities prevalent in ODD [26–28]. Early MRS and EEG results indicate the neurochemistry of DMN-node and abnormal indices associated with ADHD, thus there is a need for ODD+ADHD specificity (Vidor et al. 2024). Data from genomic and cohort studies indicate that co-occurrence of ODD results from increased polygenic load for externalizing behavior and gene–environment interaction [29–31].

Despite these advances, the literature seldom emphasizes ODD+ADHD as a primary analytical focus; numerous reviews conflate ODD with conduct disorder or examine ADHD without distinguishing ODD, thereby limiting mechanistic specificity and minimizing whether the identified neural, stress-axis, or immune signals are specific to comorbidity, additive, or influenced by symptom dimensions [1, 26, 32]. Furthermore, variability in tasks, preprocessing, comorbidity management, and covariate control, coupled with insufficient consideration of development, sex, medication, and adversity, compromises the comparability and generalizability of studies [32–34]. Dimensional research indicates that the irritable and defiant/vindictive aspects of ODD demonstrate distinct correlations with internalizing and externalizing trajectories, implying that comorbidity-specific biomarkers may align with symptom dimensions rather than syndromic classifications [35–37]. Developmental and familial-context studies reveal bi-directional coercive cycles between parents and children, likely affecting stress-related immune physiology and neural regulatory systems. This highlights the need to incorporate multilevel moderators in ODD+ADHD analyses [38, 39]. These gaps necessitate a systematic review focused on children and adolescents with ODD comorbid with ADHD, incorporating multimodal MRI (structural, task-based, and resting-state connectivity), electrophysiology, HPA-axis assessments, inflammatory biomarkers, and genetic findings. It makes a clear distinction between ODD+ADHD and groups with only ADHD or ODD/CD. It also looks at moderators (age, sex, medication, adversity) and dimensional mappings (irritability vs. defiance; cool vs. hot EF) to move away from the idea of comorbidity and toward

mechanism-informed nosology and intervention targets [6, 40].

Although ADHD has undergone many behavioral and cognitive studies, relatively fewer studies have addressed the neurobiological differences of comorbid ODD. This research gap is of significance since ODD is commonly co-occurring with an ADHD affecting 30–60% of children diagnosed with ADHD. Their combine association is linked to higher functional impairment, poorer psychosocial outcomes and enhanced risk of antisocial behavior compared with an ADHD alone [41–43]. Neurobiological studies suggest that the main features of ADHD are dysfunctions in fronto-striatal and fronto-cerebellar neural circuits, which are responsible for executive functioning, attention regulation, and inhibitory control [44, 45]. In contrast, disruptive behavior disorders like ODD and CD have been associated with abnormalities in emotios' processing and regulation, especially those involving the amygdala and prefrontal cortex [1, 46]. However, relatively few studies have compared directly the neural mechanisms underlying an individual with an ADD alone and ADD with comorbid ODD. Understanding these distinctions is important because the presence of ODD has important and significant implications for developmental trajectories and treatment response in ADHD among these populations [47].

The objective is to identify neurological and biological correlates that are shared, unique, or synergistic in ODD+ADHD compared to single-diagnosis counterparts, while assessing their developmental and contextual influence. The goals are fourfold: to describe structural, functional, and network-level brain changes that set ODD+ADHD apart from ADHD-only, ODD/CD-only, and typically developing groups; to combine electrophysiological and stress-immune biomarkers in ODD+ADHD and measure moderators like age, sex, adversity, and medication; to look at how irritability and defiance map onto certain neural circuits (like amygdala–mPFC coupling and DMN–frontoparietal interactions) and stress-immune indices; and to suggest methodological standards for future research (like comorbidity-stratified analyses, harmonized tasks and preprocessing, and preregistration) that can help clarify mechanistic inference and lead to targeted, mechanism-informed interventions like emotion-regulation training and stress-axis or inflammation-modulating adjuncts.

Methodology

Design and reporting framework

This systematic literature review (SLR) was planned and documented in accordance with PRISMA 2020 guidelines [48]. A PICO framework organized the inquiry, eligibility standards, and data documentation [49]. Due to the variability in designs and outcome metrics, the findings were integrated narratively through an inductive, reflexive thematic analysis utilizing Braun and Clarke's six-phase method [50]. Risk of bias (RoB) was evaluated using design-appropriate instruments (refer to "Risk of Bias Assessment"), and RoB determinations were incorporated into the synthesis weighting [51].

Though, meta-analysis is strong quantitative synthesis approach, this is not suitable in studies where heterogeneity in methodologies, population, sample and outcome measures. Here, statistical pooling can have misleading results, and instead, a qualitative or narrative synthesis should be used [14]. Thus, these circumstances led us to qualitative synthesis approach.

Review question and PICO

Population (P). Children and adolescents (generally aged 5–18 years) diagnosed with ADHD, with or without concurrent ODD. Research that included both children and adolescents was acceptable if outcomes were presented for the overall youth cohort or categorized by age group.

Intervention/Exposure (I). The presence of ODD comorbidity in adolescents with ADHD and/or ADHD-related interventions, where ODD status was assessed (e.g., stimulant exposure) and biological or neurological correlates were evaluated as outcomes.

Comparator ADHD absent ODD, typically developing controls, or intra-subject pre–post comparisons for intervention studies.

Results (O). The main outcomes were biological and neurological links between ADHD and ODD, both with and without ODD. Biological correlates encompassed neuroendocrine function (e.g., hair/salivary cortisol), immune/inflammatory markers (e.g., interleukins; tryptophan–kynurenine metabolites), and additional peripheral biomarkers. Neurological correlates encompassed brain structure (e.g., cortical/subcortical volumes; cerebellar metrics), brain function and connectivity (resting-state/task fMRI, DTI, fNIRS/fNIRS-hyperscanning), and electrophysiology (e.g., ERP/RewP). Neurocognitive indices intimately associated with neural systems (cool/hot executive functions, affective decision-making, temporal processing) were incorporated as secondary correlates in the analysis of biological and neurological outcomes.

The review question was: How does comorbid ODD relate to biological and neurological correlates in children and adolescents with ADHD?

The broad scope of populations was managed through transparent study selection following PRISMA 2020 guidelines and by conducting risk-of-bias assessments for all included studies to ensure balanced interpretation of the evidence [48].

Information sources and search

From the beginning until August 2025, thorough searches were done in MEDLINE (Ovid), EMBASE, CINAHL, PsycINFO, Web of Science, and Scopus. Search strings integrated controlled vocabulary and keywords pertaining to ADHD, ODD, youth terminology (child*, adolescen*, youth), and outcome domains (cortisol, interleukin*, cytokine*, kynuren*, fMRI, DTI, rs-fMRI, fNIRS, ERP, executive function*, working memory, reward, irritab*, emotional lability). There were no plans for grey literature, to maintain methodological rigor by focusing only on peer-reviewed empirical studies, ensuring higher quality and reliability of the evidence synthesis [48]. In addition to database searches, we also used backward and forward citation chasing and manually checking reference lists, conducted using Google Scholar and other citation tools by two independent reviewers, applying the same predefined inclusion criteria to ensure a comprehensive and unbiased literature search. The PRISMA diagram shows the yield for the review: We found 1,457 records in databases: MEDLINE (302), EMBASE (311), CINAHL (223), PsycINFO (287), Web of Science (198), and Scopus (136)0.37 more records from citation chasing or manual sources as shown in Fig. 1.

Eligibility criteria

Inclusion: (a) peer-reviewed empirical studies; (b) participants were children/adolescents with ADHD; (c) ODD was measured or samples were stratified by ODD status, or ODD symptoms were modeled; (d) at least one biological or neurological correlate (as defined above) was an outcome; and (e) observational (cross-sectional, case–control, cohort) or experimental/quasi-experimental designs.

Exclusion: Samples restricted to adults; lacking focus on ADHD/ODD or insufficient data on ADHD/ODD; non-empirical works (reviews, editorials, protocols without results); not subjected to peer review; case reports/series with $n < 10$; conference abstracts; studies devoid of biological/neurological outcomes.

Study selection and screening

Rayyan was used to import references for deduplication and dual-reviewer screening [52], conducted title, abstract and full-text screening inter-rater reliability Cohens' Kappa (k = 0.85), showing strong agreement and ensuring consistency in selection process. After getting rid of duplicates, there were 1,109 unique records left. Two reviewers looked at the titles and abstracts separately. If there were any disagreements, they talked about them and a third reviewer stepped in if necessary.

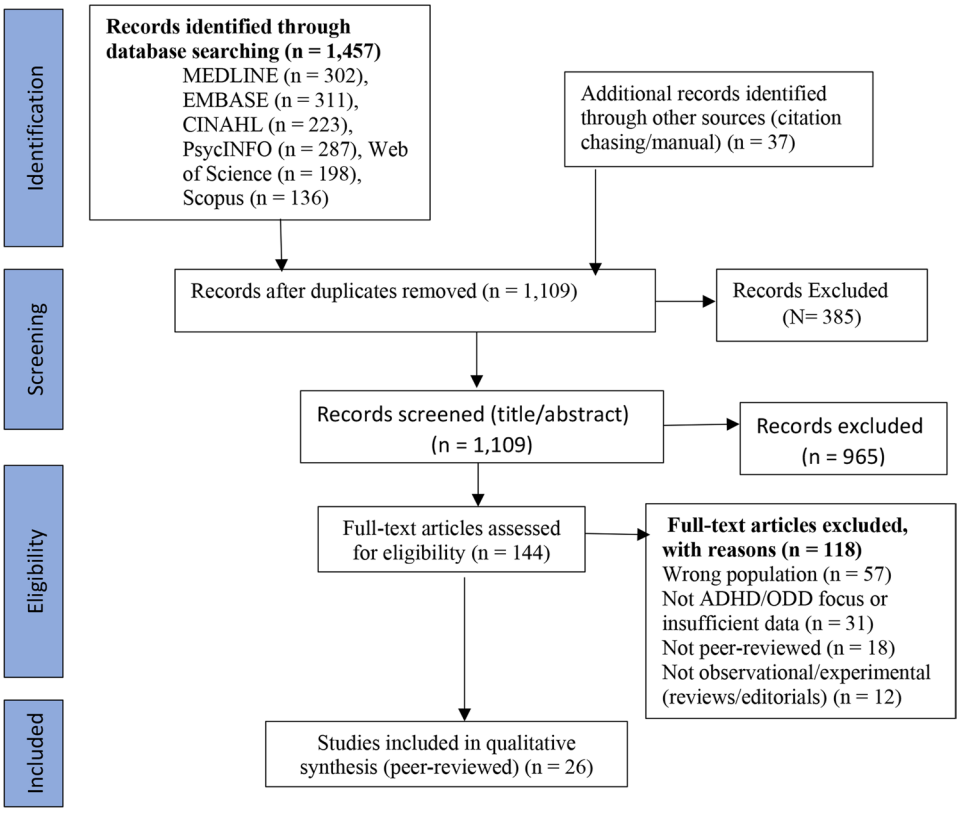


Fig. 1 PRISMA diagram

965 records were not included because of their title or abstract due to strict predefined inclusion criteria, leading to the inclusion peer-reviewed studies directly addressing neurological, neurobiological and neurocognitive correlates of ADHD and ODD comorbidities, ensuring relevance. Two reviewers looked at the full texts of 144 records to see if they met the criteria. 118 were excluded for the following reasons: incorrect population ($n = 57$), lack of ADHD/ODD focus or insufficient data ($n = 31$), absence of peer review ($n = 18$), and failure to be observational/experimental ($n = 12$). The final qualitative synthesis comprised 26 peer-reviewed studies examining biological and/or neurological correlates in child and adolescent ADHD, both with and without ODD.

Data were synthesized qualitatively because of the heterogeneous nature of included studies i.e., with different research methods, including fMRI, vMorphometry (VBM), resting-state functional near-infrared spectroscopy (fNIRS ALFF), serum tryptophan-kynurenine profiles, cytokine panels, and executive function (EF) profiling. We chose this in order to provide a comprehensive overview of the available evidence regardless the differences in designs, sampling populations and measurement methods. It is worth mentioning that qualitative synthesis provides a lower level of evidence than a quantitative strategy, such as meta-analyses, would, in which studies with the same methodology and results would need to be used. Hence, it creates the need to do careful in interpretation of the findings as exploratory but not conclusive.

Data extraction

A pilot extraction template recorded: Study design; country/setting; sample characteristics (age range, sex distribution, diagnostic instruments, medication status); ODD definition (diagnosis vs. symptom scales) and grouping; comparator(s); biological outcome details (assay type, specimen, timing, pre-analytic controls); neuroimaging/EEG acquisition and preprocessing (scanner/model, sequence, motion control, head motion thresholds, multiple-comparison corrections); neurocognitive task parameters; covariates/confounder control (e.g., age, sex, IQ, SES, adverse childhood experiences, motion); statistical models and effect estimates with confidence intervals; and funding/COI. One reviewer took out all the entries, and a second checked them.

Risk of bias assessment

Two reviewers used design-specific tools on their own. RoB 2 was used in randomized and quasi-randomized studies [53]. Non-randomized intervention and analytical observational studies utilized ROBINS-I (Sterne et al., 2016). The Joanna Briggs Institute checklist [54] was used to evaluate purely cross-sectional analytic designs. Domain-level and overall RoB judgments informed the synthesis by guiding the interpretive weighting of findings, with greater confidence placed on studies with low-to-moderate RoB, while results from higher-risk studies were interpreted more cautiously. When feasible, sensitivity analyses highlighted findings from studies with lower RoB and adequate control for key confounders such as age, sex, IQ, medication status, and motion.

Synthesis approach

Meta-analysis was not planned due to the heterogeneity of populations, exposures, and outcome metrics. The review followed Braun and Clarke's method: (1) getting to know the corpus; (2) coming up with initial codes based on biological and neurological concepts (for example, HPA-axis profiles, cytokine/tryptophan pathways, amygdala/cerebellar morphometry, cortico-striatal–cerebellar connectivity, reward-related ERPs, executive-control performance); (3) developing themes across studies while keeping in mind age band (child vs. adolescent), ODD status, and assay/modality differences; (4) reviewing themes against study quality and context; (5) defining and naming final themes; and (6) writing an analytic narrative that connects signals from lower-risk studies and explains differences by design, measurement, or confounding [50]. When authors provided effect sizes or model coefficients, these were extracted to facilitate structured, tabular comparisons across modalities and age strata.

Managing review-level bias and deviations

Before database queries, the protocol laid out plans for eligibility, screening, and analysis. We used dual-reviewer processes for screening, RoB, and extraction. Reasons for exclusion were documented and are illustrated in the PRISMA diagram [48]. Grey literature was deliberately omitted to preserve the integrity of peer-review quality control; the possibility of publication bias is addressed in the Discussion. A planned quantitative synthesis for executive-function outcomes was not conducted due to incompatible measures and statistics, as stipulated by the Cochrane Handbook [51].

Characteristics of included studies

Across the 26 sources, nine studies employed cross-sectional neurobiological or neurocognitive case–control designs to compare ADHD-only, ADHD + ODD (and occasionally ADHD + CD), and healthy controls, primarily during childhood/adolescence. These studies included amygdala-seeded rsfMRI/DWI with caregiver ratings [55], structural morphometry in adolescent boys [12], cerebellar VBM with resting-state connectivity [13], resting-state fNIRS ALFF [56], serum tryptophan–kynurenine profiling [57], cytokine panels pre/post

methylphenidate in prepubertal ADHD with screening for ODD/CD [58], matched adolescent EF profiling spanning cool/hot EF and timing [59], hot/cold EF in medication-naïve children [60], and a school sample contrasting ADHD with matched controls on ODD/CD/anger [61]. Two additional cross-sectional clinic samples examined family and temperament correlates [62], and clinical executive function and irritability differences among DMDD, ADHD, ODD, and DMDD+ADHD, utilizing both performance-based and everyday executive function measures [63]. Two longitudinal studies were conducted: a one-year cross-lagged panel network analysis involving high-risk Chinese children with ODD and/or ADHD [64], and a follow-up study from preschool to age 8 examining reward-related inhibitory control, approach motivation, and autonomic reactivity as predictors of subsequent ADHD, while considering ODD and callous–unemotional traits [65]. Two experimental/physiological challenge paradigms were incorporated: a naturalistic, single-dose methylphenidate attention task in drug-naïve ADHD individuals with and without ODD [66], and an ERP guessing task assessing reward positivity with modeled interactions among CD/ODD symptoms, irritability, and CU traits [67]. Three extensive neuroimaging/modeling studies utilized ABCD baseline data: a bifactor versus alternative factor modeling of externalizing behaviors using emotional n-back fMRI [68], a resting-state fMRI sample-entropy analysis of cool/hot executive function networks in individuals without comorbidities of ADHD and ODD and in those with comorbid presentations [69], and an amygdala-volume brain-chart methodology correlating subcortical volumes with ADHD risk and symptom severity, excluding comorbidities such as ODD [70]. A study in adolescent psychoneuroendocrinology examined the association between dimensional ADHD and diurnal cortisol, incorporating prevalent comorbid symptoms and, in sensitivity analyses, replacing ODD with CD [71]. A clinical psychology study compared ADHD with ADHD + CDS and ADHD + ODD regarding family, emotion regulation, and executive function indices [72]. The collection also included a translational editorial/review that summarized the structural findings from the ENIGMA scale and the task-fMRI effects linked to stimulants, along with notes on ODD/CD [73], as well as a registered mixed-methods pharmacokinetic protocol for supervised cannabis as an adjunct to stimulants in youth exhibiting ADHD and ODD characteristics [74]. In total, there were at least 13 studies that were cross-sectional case–control with biological or neurocognitive endpoints [12, 13, 19, 55–63, 71], two were longitudinal [64, 65], two were challenge/experimental [66, 67], three consisted large-scale ABCD imaging/modeling [68–70], one was an editorial synthesis [73], one a mixed-methods protocol [74], along with one a clinic-based EF/irritability comparison including ODD as a diagnostic cell [63].

There were clear sampling frames and group definitions in more than twenty studies, and most of them had three or four arms. A large clinical study compared 2,015 children with ADHD to 745 typically developing controls in terms of emotional lability and monitored the developmental trajectory of ADHD symptoms and oppositional defiant disorder (ODD) symptoms [8]. Imaging cohorts were stratified by modality and age: rsfMRI/DWI comprised 354 ADHD-only participants, 161 ADHD + ODD participants, and 100 controls, with an imaging subsample of 209 [55]; VBM/rsFC analyses utilized ADHD-200 subsamples of 27 ADHD-only, 27 ADHD + ODD, and 27 controls aged 9–14 years [13]; adolescent morphometry matched 36 ADHD-only boys, 26 ADHD + ODD/CD boys, and 30 typically developing boys aged 11–17 [12]; resting-state fNIRS included 165 medication-naïve children aged 7–15 years across ADHD, ADHD&LD, ADHD&ODD, and controls [56]; hyperscanning fNIRS consisted of 72 mother–child dyads, 35 ODD and 37 healthy controls, during free-play and valenced discussions [64]. Biological assays were conducted in prepubertal or medication-free cohorts: serum kynurenine metabolites in 46 ADHD-C, 43 ADHD-C + ODD, 33 ADHD-C + CD, and 50 controls [57]; cytokines at 09:00 before and after ~4.6 months of methylphenidate in 31 prepubertal inattentive vs. combined ADHD screened for "ODCD" and anxiety [58]; hair cortisol and skin-conductance in 98 medication-naïve ADHD and 107 healthy controls with DSM-5 comorbidity assessment [19]. The methylphenidate challenge included 53 drug-naïve individuals with combined-type ADHD (26 with ADHD + ODD and 27 without ODD), all possessing an IQ of 80 or higher and excluding other neurodevelopmental or psychiatric disorders [66]. The ABCD-based samples were substantial: 11,878 9-year-olds for the bifactor/task-fMRI model (Perlstein et al., 2024) and 61 individuals with comorbid-free ADHD, 38 with ODD, 48 with OCD, 833 with comorbid presentations, and 269 matched controls for the rsfMRI-complexity study [69]. The cross-lagged panel network monitored 263 high-risk children aged 6 to 13 years over the course of one year, creating two cross-sectional networks and a CLPN [64]. A community school sample was utilized to match ADHD and non-ADHD groups by sex (50 boys and 58 girls per group; ages 6–15) for the purpose of comparing ODD/CD/anger [61]. A clinic-based developmental cohort commenced at preschool (mean age 58 months) and was re-evaluated at approximately 8 years (mean age 101.4 months) for ADHD outcomes, while also documenting ODD/CU symptoms [65]. A child psychiatry outpatient sample examined DMDD, ADHD, ODD, and DMDD+ADHD in relation to everyday executive

function (EF) versus task EF and irritability [63]. A clinic/community sample of 8–12-year-olds ($n = 842$) compared ADHD, ADHD + ODD, ADHD + CDS, and controls on family and executive function indices [72]. A sample of at-risk adolescents ($N = 140$; 38% female) was utilized to evaluate the standardization of the brain-chart amygdala [70]. Lastly, the mixed-methods protocol focused on 10 to 20 adolescents aged 12 to 25 who exhibited ADHD and ODD characteristics and were currently utilizing a THC: CBD 1:20 herbal extract in conjunction with stimulants, detailing pharmacokinetic sampling and validated symptom scales [74].

The modalities and measures encompassed structural and functional neuroimaging, neurophysiology, cognitive and behavioral phenotyping, and peripheral biology. Structural MRI with FreeSurfer produced indices for total gray matter, surface area, and cortical thickness, identifying a thicker right rostral middle frontal cortex in individuals with ADHD and ODD/CD [12]. VBM identified diminished anterior cerebellar gray matter volume (Cerebellum_4_5) exhibiting hyperconnectivity to executive function and default mode network regions in individuals with ADHD and ODD [13]. Amygdala-seeded resting-state functional MRI (rsfMRI) and diffusion-weighted imaging (DWI) measured subnetworks and streamline density in conjunction with the Child Behavior Checklist (CBCL) and the Behavior Rating Inventory for Executive Function (BRIEF) [55]. Resting-state fNIRS generated ALFF matrices across prefrontal and temporal channels, indicating elevations in ADHD and LD, with no supplementary effect of ADHD and ODD beyond ADHD [56]. Hyperscanning fNIRS measured phase-locked synchrony in the right MFG/SFG, left TPJ/SFGdor, and bilateral SFGmed/right ANG/left MFG, revealing diminished synchrony in ODD dyads and negative correlations with ODD symptoms [75]. Task fMRI during emotional n-back connected a general externalizing factor to lower right amygdala reactivity to fearful faces and linked ADHD scores to higher right nucleus accumbens responses to fearful/happy faces [68]. Resting-state fMRI complexity, assessed through voxel-wise sample entropy, revealed concurrent reductions in the bilateral SFG, ACC/PCC, and caudate in individuals with comorbidity-free ADHD and ODD, with more extensive reductions observed when ADHD coexisted with additional disorders [69]. ERPs assessed reward positivity approximately 300 ms post-feedback, indicating a CD × irritability interaction [67]. Peripheral biology encompassed LC–MS/MS quantification of kynurenine pathway metabolites and ratios [57], multiplex cytokine assessments at baseline and after approximately 4.6 months of methylphenidate treatment, with interactions analyzed by ADHD presentation and "ODCD" comorbidity [58], two-day salivary cortisol profiles measuring CAR and AUCg while controlling for CD/anxiety/depression and substituting ODD in sensitivity analyses [71], and hair cortisol/skin conductance as indicators of HPA activity and sympathetic reactivity, stratified by sex, ADHD presentation, and ODD/CD [19]. Behavioral phenotyping utilized the CBCL/BRIEF [55], Disruptive Behavior Disorders Scale and Beck Youth Anger Inventory [61], multi-informant ODD/CU assessments [65], SNAP-IV/PHQ-9/SCARED within the cannabis protocol [74], and comprehensive EF batteries encompassing inhibition, working memory, reinforcement processing, emotion recognition, and time production/reproduction [59], alongside WAIS-R Digit Span, Corsi Block, WCST, delay discounting, and Iowa Gambling [60]. Family context and temperament were evaluated through parental affective temperaments, parental ADHD and internalizing symptoms, parenting styles, and child anxiety/depression/autistic traits in relation to ODD/CD severity [62]. In a cohort of 8–12-year-olds, the assessment included family dynamics, emotion dysregulation, social exclusion/solitude, and CNSVS [72]. Emotional lability was assessed using ROC analyses and developmental patterns according to diagnostic and comorbid status [8]. The translational review contextualized these modalities within ENIGMA-level subcortical reductions and stimulant-associated normalization of inhibitory networks, highlighting amygdala findings and associations with ODD/CD from previous syntheses [73].

Analytic strategies corresponded with modality and design. ANCOVA examined caregiver-assessed disparities in behavioral executive function [55]. One-way ANOVA assessed ALFF channel disparities [56]. Voxel-wise group comparisons were utilized for rsfMRI/DWI [55] and VBM/connectivity [13]. FreeSurfer-based surface/volume models differentiated diagnostic groups and correlated thickness with symptom severity [12]. Hierarchical GEE with Benjamini–Hochberg corrections analyzed sample entropy across EF networks [69]. The ABCD data were analyzed using bifactor, four-correlated, and second-order models, with the bifactor solution most accurately depicting shared general externalizing and specific CD, ODD, ADHD, and CU factors [68]. Cross-lagged panel networks discerned stable bridges ("annoy," "blame") and reciprocal/unidirectional paths over one year [64]. Factorial analyses examined cytokine presentation-by-comorbidity interactions, revealing significant effects of IL-1β and IL-13 in inattentive ADHD that were eliminated following treatment [58]. Mediation associated enhanced inhibition at elevated CD symptoms via SMN(L)–DMN(R) connectivity in younger children [8]. ERP models identified a CD × irritability interaction on RewP [67]. ROC exhibited significant screening effectiveness of emotional lability for ADHD [8]. The methylphenidate challenge compared reaction-time enhancements

between ADHD without ODD and ADHD with ODD, utilizing rigorously matched baselines [66].

Lastly, five studies explicitly examined developmental and sex stratifications. Emotional lability diminished with age in typically developing children but continued in those with ADHD, whereas ODD symptoms increased developmentally within the ADHD population [8]. Externalizing symptoms were more pronounced in boys compared to girls, with no significant age effect observed in a school sample [61]. Hair-cortisol and sympathetic reactivity exhibited sex-by-presentation effects, alongside diminished skin conductance in boys with ADHD + ODD/CD; controlling for adverse childhood experiences eliminated hair-cortisol disparities, yet sympathetic blunting persisted [19]. The adolescent morphometry study was intentionally conducted with a male-only cohort [12]. The brain-chart amygdala study encompassed both sexes (38% female) and revealed amygdala volume correlations with ADHD risk and severity along with comorbidities including ODD [70]. Collectively, these characteristics represent a methodologically diverse yet rigorously specified evidence base connecting neural systems, neuroendocrine/autonomic function, immune–metabolic markers, executive phenotypes, and family/contextual moderators to ODD comorbidity within ADHD in youth [8, 12, 13, 19, 56–75]. The demographic representation of the studies are shown in Fig. 2. While the risk of bias analysis shown in Tables 1 and 2 is the summary of the studies included.

Thematic analysis

Theme 1: limbic–striatal signatures, amygdala metrics, and reward responsivity

In six studies, limbic–striatal structure and function consistently demonstrated relevance to oppositionality in ADHD, with several elucidating how comorbidity influences these neural signatures [55, 67–70, 73]. In a comprehensive multimodal study comprising 354 children with ADHD only, 161 children with ADHD and ODD, and 100 control subjects (including a 209-participant imaging subsample), the ADHD-only group exhibited enhanced functional connectivity between amygdala subregions and the left caudate, left putamen, and frontal cortex compared to both control and ADHD + ODD groups. Additionally, the ADHD-only group demonstrated increased structural streamline density in the left internal capsule, corpus callosum, and right superior corona radiata. These alterations in structural connectivity are associated with emotional difficulties, while changes in functional connectivity are linked to ADHD-related characteristics [55]. Utilizing brain-chart reference standards in adolescents ($N = 140$), a reduced

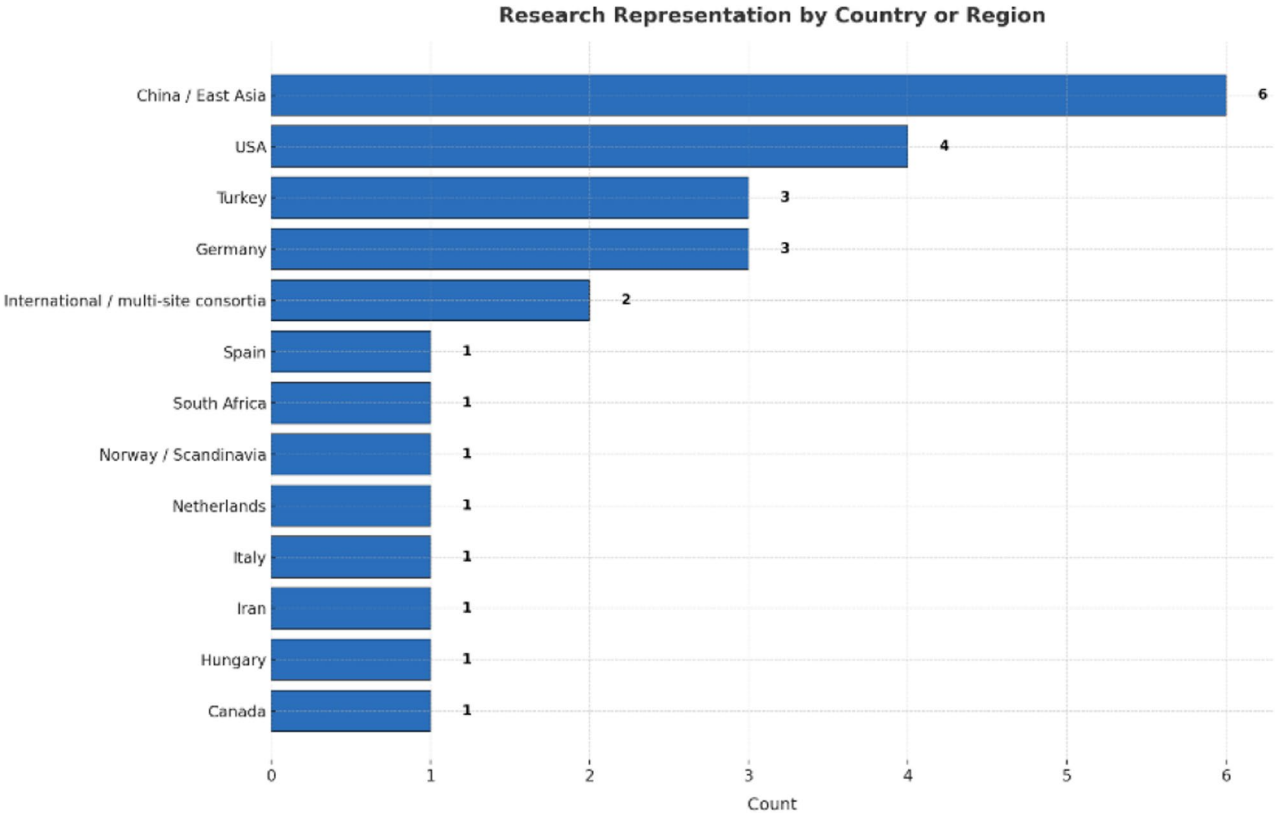


Fig. 2 Geographical representation of studies ($n = 26$)

Table 1 Risk-of-bias analysis

Study #	Citation	Design / Population	Selection Bias	Confounding Control	Overall Risk of Bias (adapted ROBINS-I)
1	[55] (BMC Psychiatry)	Case–control; ADHD-only ($n=354$), ADHD + ODD ($n=161$), HC ($n=100$); imaging subsample $n=209$	Moderate — large behavioral sample but imaging on a subsample may be non-representative; clinic-based recruitment likely.	Moderate — ANCOVA; Bonferroni for behavior; unclear motion/scan QC or covariates beyond basic demographics.	Moderate — strong sample size; subsample imaging and incomplete measurement/QC details.
2	[8] (J Atten Disord)	Large cross-sectional validation; ADHD ($n=2{,}015$), TDC ($n=745$)	Low — very large, well-defined cohorts reduce spectrum/selection concerns.	Low — examined sex, IQ, age; ROC analyses; comorbid ODD accounted; robust concurrent validity approach.	Low — strong sampling and analytical controls.
3	[72] (J Atten Disord)	Cross-sectional; ages 8–12; ADHD ($n=246$), ADHD + ODD ($n=212$), ADHD + CDS ($n=176$), Controls ($n=207$)	Moderate — sizable groups but clinic-based; potential referral bias.	Moderate — multiple psychosocial/EF measures; abstract does not specify covariate adjustments for SES/family factors.	Moderate — breadth of measures; limited confounding detail.
4	[63] (Eur Child Adolesc Psychiatry)	Clinical comparison; DMDD, ADHD, ODD, DMDD+ADHD; ages 6–12; standardized interviews	Moderate — referred clinical sample; representativeness uncertain.	Moderate — dual EF modalities; irritability modeled; covariate set not detailed.	Moderate — sound diagnostics; limited adjustment reporting.
5	[64] (Eur Child Adolesc Psychiatry)	Longitudinal high-risk; $n=263$; two waves; CLPN networks	Low — repeated assessment in defined high-risk cohort; acceptable retention implied.	Low — cross-lagged panel networks address temporal precedence; bridge symptoms modeled; residual time-varying confounding possible.	Low — appropriate longitudinal modeling for symptom dynamics.
6	[66] (PLOS ONE)	Naturalistic, pre–post single-dose MPH; ADHD w/ and w/o ODD; $n=53$, drug-naïve	Moderate — selective clinic cohort; small sample; first-dose responders only.	Moderate — groups balanced on demographics/attention; behavioral dysregulation differences may confound MPH response.	Moderate — careful inclusion/exclusion; limited power and generalizability.
7	[57] (Psychiatry Res.)	Cross-sectional biomarkers; ADHD-C ($n=46$), ADHD-C + ODD ($n=43$), ADHD-C + CD ($n=33$), HC ($n=50$)	Moderate — medication-free is a strength; sample sizes modest; clinic-based.	High — potential confounding by infection, BMI, diet/circadian not specified; ratios mitigate but do not eliminate confounding.	Moderate to High — analytic validity strong (LC–MS/MS); contextual confounding likely.
8	[61] (SA J Educ.)	School-based case–control; $n=216$; teachers/parents/self-reports	Moderate — matched groups; potential unmeasured school/context factors.	Moderate — multi-informant ratings; cross-sectional; limited covariate adjustment reported.	Moderate — ecologically valid ratings; residual confounding likely.
9	[62] (Children's Health Care)	Cross-sectional family/child predictors; ADHD youth; ODD/CD symptoms	Moderate — clinic-based; unclear sampling frame.	Moderate — models include parental temperament, parenting, child traits; risk of overfitting and unmeasured SES.	Moderate — rich covariates; observational limitations.
10	[59] (J Atten Disord)	Matched case–control; ADHD-only ($n=82$), ADHD + ODD ($n=82$), HC ($n=82$)	Low — matching on age, sex, IQ, ADHD type reduces selection imbalance.	Low — cool/hot EF and timing assessed; design isolates ODD effects beyond ADHD.	Low — rigorous matching and measurement.
11	[19] (Eur Child Adolesc Psychiatry)	Case–control; ADHD ($n=98$) vs. HC ($n=107$); medication-naïve; HCC & SCR	Moderate — community representativeness unclear; adequate sample.	Low — accounted for symptom presentation, ODD/CD, sex, and ACE; robust moderation tests.	Low to Moderate — strong confounding control; typical case–control limits.
12	[67] (Res Child Adolesc Psychopathol.)	ERP RewP; ADHD with conduct problems; $n=48$, treatment-referred	Moderate — small, referred sample; selection toward greater impairment.	Moderate to High — modeled CD×irritability; limited power for interactions; ERP QC not detailed.	Moderate to High — hypothesis-generating due to size/QC detail.
13	[73] (World J Biol Psychiatry)	Editorial/overview; no primary outcomes	N/A — not a primary study.	N/A — not a primary study.	N/A — not applicable to RoB.
14	[70] (Res Child Adolesc Psychopathol.)	Adolescent clinical sample; $N=140$; brain chart norms; amygdala volume vs. ADHD risk/severity	Moderate — modest sample; clinic-based.	Moderate — adjusts for anxiety, depression, ODD symptoms, and motion; residual confounding possible.	Moderate — principled normalization; limited size.

Table 1 (continued)

Study #	Citation	Design / Population	Selection Bias	Confounding Control	Overall Risk of Bias (adapted ROBINS-I)
15	[74] (JMIR Res Protocols)	Mixed-methods PK/effectiveness protocol; ADHD with ODD features	N/A — protocol only.	N/A — protocol only.	N/A — not applicable until results.
16	[13] (Int J Dev Neurosci.)	ADHD-200 repository; ADHD-only ($n=27$), ADHD + ODD ($n=27$), HC ($n=27$); VBM & rs-FC	Moderate — balanced small groups; public dataset limits selection bias but small n.	Moderate to High — motion/QC and multiplicity not detailed; many FC tests.	Moderate — consistent signals; constrained by size/QC detail.
17	[75] (NeuroImage)	Dyadic fNIRS hyperscanning; mother–child; ODD ($n=35$) vs. HC ($n=37$)	Moderate — convenience dyads; referral bias possible.	Moderate — task variants; synchrony clustering; covariates not detailed (e.g., SES/parental psychopathology).	Moderate — novel measure; small n; limited confounder reporting.
18	[69] (medRxiv)	ABCD rs-fMRI complexity; comorbid-free ADHD ($n=61$), ODD ($n=38$), OCD ($n=48$), comorbid ($n=833$), controls ($n=269$)	Low — multisite population-based sampling within ABCD.	Low — GEEs; BH correction; explicit comorbidity contrasts across EF networks.	Low — strong design and multiplicity control; preprint status noted.
19	[68] (J Psychopathol Clin Sci)	ABCD emotional N-back fMRI; $N \approx 11{,}878$; bifactor and alternative models	Low — large, multi-site cohort; high representativeness for age group.	Low — latent modeling partitions general vs. specific variance; multiple model specifications.	Low — robust psychometric and imaging framework.
20	[56] (Alpha Psychiatry)	Resting fNIRS; 4 groups (ADHD, ADHD&LD, ADHD&ODD, HC); $n=165$; medication-naïve	Moderate — decent size; clinic-based; Chinese sample.	Moderate to High — ALFF metric; covariate controls (e.g., symptom severity, motion) not specified.	Moderate — informative but QC/adjustment details limited.
21	[76] (Asian J Psychiatry)	Medication-naïve ADHD with ODD vs. CD vs. HC; EF & FC mediation	Moderate — sizable clinical sample; representativeness unclear.	Moderate — mediation of FC between CD symptoms and inhibition; broader confounders not detailed.	Moderate — thoughtful modeling; observational constraints.
22	[58] (Biomedicines)	Pre–post MPH cytokines; ADHD-AD ($n=13$) vs. ADHD-C ($n=18$); screen for ODCD/anxiety	High — very small, prepubertal sample; possible selection on treatment initiation.	High — multiple cytokines; limited correction; comorbidity interactions noted but many uncontrolled influences.	High — exploratory; limited precision.
23	[65] (Front Psychiatry)	Prospective cohort; preschool to ~8 years; $n=198$; multi-informant	Low to Moderate — developmental cohort; some attrition possible.	Low to Moderate — models include ODD/CU; RRIC and EDA; potential residual family/SES confounding.	Moderate — longitudinal strength; observational limits.
24	[71] (Psychoneuroendocrinology)	Adolescents ($n=138$); 2-day saliva; dimensional psychopathology	Moderate — volunteer sample; compliance variability not detailed.	Moderate — controlled for CD, anxiety, depression; sensitivity with ODD; adherence/assay precision not in abstract.	Moderate — solid modeling; measurement detail limited.
25	[12] (Hum Brain Mapp.)	Structural MRI; boys only; ADHD-only ($n=36$), ADHD + ODD/CD ($n=26$), TD ($n=30$)	Moderate — small male-only sample limits generalizability.	Moderate — controlled for age/puberty/IQ; thickness linked to ODD/CD symptoms; other confounders possible.	Moderate — careful morphometry; limited size/sex scope.
26	[60] (Int J Dev Neurosci.)	Cross-sectional EF tasks; ADHD w/ ODD ($n=15$), ADHD ($n=15$), TD ($n=15$); medication-naïve	High — very small groups; clinic-based.	High — multiple EF tests; limited covariate control; multiplicity risk.	High — exploratory; underpowered.

Graphical Representation

Table 2 Summary of included studies

Authors (Year)	Journal	Aims	Methodology	Findings	Limitations
[55]	BMC Psychiatry	To investigate structural and functional connectivity of amygdala subregions in children with ADHD-only, ADHD + ODD, and controls, and relate these to behavioral/emotional problems.	Cross-sectional; 354 ADHD-only, 161 ADHD + ODD, 100 controls; CBCL and BRIEF with ANCOVA; subsample ($n = 209$) had rs-fMRI and DWI; voxel-wise FC and SC analyses using amygdala subregions as seeds.	ADHD + ODD showed more anxiety/depression, delinquent and aggressive behavior, emotional control issues, and inhibition deficits; ADHD-only showed increased amygdala FC with left caudate, putamen, and frontal cortex and higher streamline density in several white matter tracts; altered SC related to emotional problems, altered FC to other ADHD features.	Limitations not reported in abstract.
[8]	Journal of Attention Disorders	To evaluate concurrent validity, screening utility, and developmental patterns of emotional lability (EL) in children with ADHD, with and without ODD.	2,015 children with ADHD and 745 typically developing controls; EL ratings; analyses by gender, IQ, comorbid ODD, ADHD subtype, and age; ROC analyses for screening performance.	ADHD showed higher EL than controls, independent of gender and IQ; EL particularly elevated in ADHD + ODD and ADHD-combined subtype; EL had strong screening efficacy for ADHD; EL decreased with age in controls but persisted in ADHD; ODD symptoms increased with age and appeared distinct from EL.	Limitations not reported in abstract.
[72]	Journal of Attention Disorders	To compare family functioning, emotion regulation, social factors, psychopathology, and executive functions among ADHD, ADHD + ODD, ADHD + CDS, and controls.	842 children aged 8–12 categorized as ADHD, ADHD + ODD, ADHD + CDS, and controls; solitude/social exclusion scales, emotion dysregulation scales, Barkley SCT, CBCL, family assessment device, CNSVS neurocognitive tests.	ADHD + CDS showed more internalizing problems, preference for solitude, poorer family communication/problem solving, and more difficulty understanding others' emotions; ADHD + ODD had poorer CNSVS performance on most domains; ADHD + CDS had lowest psychomotor speed and poorer reaction time and cognitive flexibility than pure ADHD.	Limitations not reported in abstract.
[63]	European Child & Adolescent Psychiatry	To compare executive function in DMDD vs. ADHD vs. ODD, and examine associations between irritability and executive function in a clinical sample.	Clinically referred children (6–12 years); parent-rated daily-life EF and performance-based neuropsychological tasks; standardized semi-structured diagnostic interview; parent-rated irritability; comparisons across DMDD, ADHD, ODD, DMDD+ADHD and dimensional analyses.	In daily life, DMDD showed worse emotion control than ADHD and elevated cognitive flexibility problems vs. norms; DMDD had fewer working memory problems than ADHD; no differences between DMDD and ODD; higher irritability was associated with emotional dyscontrol and cognitive inflexibility; no diagnostic or irritability-related differences emerged on performance-based EF measures.	Limitations not reported in abstract.
[64]	European Child & Adolescent Psychiatry	To examine stability of ODD–ADHD comorbidity symptom networks over time and longitudinal relations between ODD and ADHD symptoms at the symptom level.	263 Chinese children (6–13) with ODD and/or ADHD; symptoms measured twice, 1 year apart; two cross-sectional networks and a cross-lagged panel network constructed for ODD and ADHD symptoms.	Network structure was highly stable across time; symptoms such as "follow through," "interrupts/intrudes," "difficulty playing quietly," and "concentration" were most central; "annoy" and "blame" acted as bridge symptoms between ODD and ADHD; "vindictive" predicted multiple ADHD symptoms, whereas "angry" was predicted by ADHD symptoms; results support broad bidirectional predictive links and partial applicability of the developmental precursor model at symptom level.	Limitations not reported in abstract.
[66]	PLOS ONE	To examine whether a single dose of methylphenidate (MPH) has different effects on attention in ADHD with vs. without comorbid ODD, and which clinical features relate to MPH response.	Naturalistic study; 53 drug-naïve children/adolescents with ADHD combined type (26 ADHD + ODD, 27 ADHD-only); at least 6 months of psychosocial/behavioral intervention; single-dose MPH; attention performance (reaction times) pre/post; baseline severity, impairment, and behavioral/emotional symptoms assessed.	MPH improved attention (reduced reaction times) more in ADHD-only than ADHD + ODD; groups did not differ in baseline attention, demographics, ADHD severity, or adaptive behavior; groups differed in behavioral profiles such as Dysregulation Profile and disruptive behaviors; combining symptom assessment with ADHD diagnosis may help identify those who benefit most from MPH.	Limitations not reported in abstract.

Table 2 (continued)

Authors (Year)	Journal	Aims	Methodology	Findings	Limitations
[57]	Psychiatry Research	To investigate the role of the tryptophan metabolic pathway (TMP) in ADHD combined type with and without ODD or CD.	46 ADHD-C alone, 43 ADHD-C + ODD, 33 ADHD-C + CD, 50 healthy controls; serum levels of tryptophan, kynurenine, kynurenic acid, 3-hydroxykynurenine, 3-hydroxyanthranilic acid measured by LC–MS/MS; enzyme-activity and neuroprotection indices calculated via metabolite ratios.	Compared with controls, kynurenine was higher in ADHD-C; 3-hydroxykynurenine was lower in ADHD-C and ADHD-C + ODD; kynurenic acid/kynurenine ratio was higher in all ADHD groups; 3-hydroxykynurenine/kynurenine ratio was lower in ADHD-C; results suggest TMP involvement in ADHD-C pathophysiology across comorbidity subgroups.	Limitations not reported in abstract.
[61]	South African Journal of Education	To determine whether children with ADHD show more ODD, CD, and anger symptoms than controls, and whether gender and age influence these comorbidities.	216 school children aged 6–15 divided into ADHD and matched control groups (50 boys, 58 girls per group); teacher and parent DBD Scale ratings; child self-report Anger Inventory.	Children with ADHD had significantly higher ODD, CD, and anger symptoms than controls; externalizing problems were more pronounced in boys with ADHD than in girls; age did not affect symptom levels; authors recommend early identification and intervention for externalizing comorbidities.	Limitations not reported in abstract.
[62]	Children's Health Care	To evaluate how parental temperament, parental psychopathology, parenting styles, and child symptoms and traits relate to ODD and CD symptoms in children with ADHD.	Cross-sectional study of children with ADHD; measured maternal and paternal affective temperaments, parental ADHD, depression, anxiety, parenting styles; child depression, anxiety, autistic traits, ADHD, OCD symptoms; examined relations with child ODD and CD symptom scores.	Maternal anxious/irritable temperament and child inattention, hyperactivity–impulsivity, and OCD symptoms were positively associated with ODD scores; parental acceptance/involvement was negatively associated with ODD scores; CD scores were positively related to maternal irritable and paternal cyclothymic/hyperthymic temperaments, and to child ODD and autistic traits, but negatively to maternal ADHD; parental affect, parenting style, and child traits may shape disruptive symptom severity.	Limitations not reported in abstract.
[59]	Journal of Attention Disorders	To assess cool and hot executive functions and temporal processing in ADHD with and without comorbid ODD.	82 ADHD-only, 82 ADHD + ODD, 82 controls (mean age 16), matched on age, gender, IQ, and ADHD type; tasks for cool EF (inhibition, working memory), hot EF (reinforcement processing, emotion recognition), and temporal processing (time production/reproduction).	ADHD + ODD showed abnormalities in inhibition, working memory, facial emotion recognition, and temporal processing; ADHD-only was impaired only in working memory and time production; results suggest that ODD accounts for a substantial part of executive-function deficits often attributed to ADHD.	Limitations not reported in abstract.
[19]	European Child & Adolescent Psychiatry	To examine long-term cortisol secretion and sympathetic reactivity in children with ADHD, considering sex, ADHD presentation, and comorbid ODD/CD.	98 children with ADHD and 107 healthy controls; DSM-5 symptom presentation and ODD/CD assessed via clinical interviews; hair cortisol concentration indexed chronic HPAA activity; skin conductance responses indexed sympathetic reactivity; internalizing symptoms and adverse childhood experiences were assessed; medication-naïve sample.	Boys with ADHD-inattentive had lower hair cortisol than healthy boys; girls with ADHD-combined had higher hair cortisol than healthy girls; boys with ADHD + ODD/CD showed blunted skin conductance reactivity; controlling for adverse childhood experiences removed cortisol but not reactivity group differences; hair cortisol appears to mark ADHD-related symptom presentation and comorbidity, and may help define more homogeneous subgroups.	Limitations not reported in abstract.
[67]	Research on Child and Adolescent Psychopathology	To test how irritability and callous-unemotional traits influence associations between ADHD/CD/ODD symptoms and reward responsivity (RewP) in youth with ADHD and conduct problems.	48 youth (5–12 years) with ADHD referred for conduct problems; EEG during a monetary guessing task; RewP to gain and loss feedback calculated; frequentist and Bayesian analyses of main effects of ADHD, CD, non-irritable ODD, and interactions with irritability and CU traits.	A robust CD × irritability interaction emerged: enhanced RewP when irritability was high and CD symptoms were low; CU traits did not moderate RR associations; little support was found for links between RR and ADHD or ODD symptoms; variability in neural reward response appears partly driven by the combination of CD symptoms and irritability.	Limitations not reported in abstract.

Table 2 (continued)

Authors (Year)	Journal	Aims	Methodology	Findings	Limitations
[73]	The World Journal of Biological Psychiatry	To review and contextualize recent translational neuroimaging findings in ADHD and related neurodevelopmental disorders, including ODD/CD and OCD, and implications for treatment.	Narrative/editorial overview of large imaging consortia (e.g., ENIGMA) and meta-analyses on structural and functional MRI, DMN connectivity, amygdala involvement, and stimulant effects; integrates multiple prior studies rather than new data collection.	ADHD is associated with small but widespread subcortical and cortical volume reductions; neuroimaging findings overlap with other disorders but show differential patterns; amygdala abnormalities may be particularly linked to ODD/CD symptoms; DMN dysfunction and its normalization by methylphenidate are emphasized; highlights the need for transdiagnostic and disorder-specific imaging approaches to inform treatment.	No empirical study; limitations of narrative synthesis and variability across underlying studies not detailed in abstract.
[70]	Research on Child and Adolescent Psychopathology	To test whether brain chart–based subcortical and total volume deviations, especially in amygdala, are associated with ADHD risk and symptom severity beyond comorbidities including ODD.	140 adolescents (38% female); brain chart reference standards applied to subcortical and total volumes; ADHD risk status and parent-rated inattention/hyperactivity, self-rated anxiety/depression, and parent-rated ODD, plus motion, entered into models.	Smaller bilateral amygdala volume was associated with ADHD at-risk status beyond comorbidities and motion; smaller amygdala volume predicted higher inattention and hyperactivity/impulsivity beyond comorbidities except ODD and motion; suggests amygdala volume meaningfully contributes to estimating ADHD risk and severity and supports use of brain charts clinically.	Limitations not reported in abstract.
[74]	JMIR Research Protocols	To design a proof-of-concept mixed-methods study on pharmacokinetics and perceived effectiveness of cannabis as an adjunct treatment for ADHD with ODD features.	Protocol; planned recruitment of 10–20 patients aged 12–25 with ADHD and ODD features using a 1:20 THC: CBD herbal extract adjunctive to stimulants; quantitative symptom scales (SNAP-IV, PHQ-9, SCARED), side-effect survey, semistructured interviews; cannabis product and two blood samples for LC–MS/MS; descriptive statistics, ANOVA, regression, and qualitative thematic analysis.	No outcome data yet; protocol outlines feasibility and analytic plan to relate cannabinoid levels to symptom change and subjective experiences in medically supervised cannabis use for ADHD + ODD.	Study is a protocol only; effectiveness and safety results not yet available.
[13]	International Journal of Developmental Neuroscience	To explore effects of comorbid ODD on gray matter volume and functional connectivity in children with ADHD.	Data from ADHD-200 repository; 27 ADHD-only, 27 ADHD + ODD, 27 healthy controls aged 9–14; voxel-based morphometry for GMV; rs-fMRI functional connectivity analyses; group comparisons.	ADHD + ODD children showed greater reduction in anterior cerebellar GMV (Cerebellum_4_5) compared with ADHD-only and controls; Cerebellum_4_5 exhibited increased functional connectivity with multiple cortical regions including EF and DMN nodes; these GMV reductions and hyperconnectivity may exacerbate emotional and cognitive deficits in ADHD.	Limitations not reported in abstract.
[75]	NeuroImage	To examine mother–child interpersonal neural synchrony (INS) during interactions and its association with children's ODD symptoms.	72 mother–child dyads (35 ODD, 37 controls); fNIRS recording from frontal, parietal, and temporal regions during free-play and positive/negative discussion tasks; synchrony assessed via phase-locked value; k-means clustering and alignment tests; network analyses linking INS patterns to ODD symptom networks.	Distinct INS patterns were observed across interaction contexts; mother–child dyads with ODD showed lower INS across frontal, parietal, and temporal regions compared to controls; INS was negatively associated with ODD symptoms; network analyses indicated INS patterns connect to different nodes within the ODD symptom network, suggesting INS as a neural marker of parent–child interaction difficulties in ODD.	Limitations not reported in abstract.

Table 2 (continued)

Authors (Year)	Journal	Aims	Methodology	Findings	Limitations
[69]	medRxiv (preprint)	To identify distinct and shared resting-state fMRI complexity (sample entropy) patterns in pre-adolescent children with ADHD, ODD, OCD, and comorbid presentations within cool and hot EF networks.	ABCD Study baseline data; 61 comorbid-free ADHD, 38 comorbid-free ODD, 48 comorbid-free OCD, 833 with comorbid ADHD/ODD/OCD/other diagnoses, and 269 matched controls; voxel-wise SampEn calculated; mean SampEn in EF network regions; hierarchical GEE models comparing diagnostic and comorbid groups.	Comorbid-free ADHD and ODD showed reduced SampEn in overlapping EF-network regions (bilateral superior frontal gyrus, anterior/posterior cingulate, bilateral caudate); ADHD additionally showed reduced complexity in right inferior/middle frontal and bilateral orbitofrontal cortex; in comorbid presentations, added ADHD symptomatology was associated with further reduced SampEn in all regions, indicating that comorbid ADHD amplifies neural complexity deficits.	Preprint; peer review and detailed methodological constraints not discussed in abstract.
[68]	Journal of Psychopathology and Clinical Science	To test structural models (bifactor vs. alternatives) of CD, ODD, ADHD, and CU traits and to identify unique vs. shared neural correlates of externalizing psychopathology.	ABCD baseline data; $N = 11,878$ children (mean age 9); questionnaire measures of CD, ODD, ADHD, CU; emotional N-back fMRI task; model comparisons among bifactor, four-correlated-factor, and second-order factor structures; brain–symptom associations examined.	A bifactor model with a general externalizing factor plus specific CD, ODD, ADHD, and CU factors fit best, though other models also fit well; across models, greater general externalizing was linked to reduced right amygdala activity to fearful faces; higher CU traits were linked to reduced dorsolateral prefrontal activation to fearful faces; higher ADHD scores were associated with greater right nucleus accumbens activation to fearful and happy faces.	Limitations not reported in abstract.
[56]	Alpha Psychiatry	To compare resting-state brain activity in medication-naïve children with ADHD, ADHD with learning disabilities, ADHD with ODD, and healthy controls using fNIRS to identify diagnostic biomarkers.	165 medication-naïve children aged 7–15 divided into ADHD, ADHD&LD, ADHD&ODD, and healthy controls; multichannel fNIRS over prefrontal and temporal regions at rest; ALFF computed; one-way ANOVA across channels.	All ADHD groups showed higher ALFF in various regions versus controls; ADHD&LD showed higher ALFF in medial prefrontal, temporal cortex, and left ventrolateral prefrontal cortex; ADHD&ODD did not differ significantly from ADHD-only; authors conclude ADHD&LD may have a distinct pathogenesis, whereas ODD symptoms may represent part of the ADHD phenotype rather than a separate disorder.	Limitations not reported in abstract.
[76]	Asian Journal of Psychiatry	To investigate differences in executive-function impairments between medication-naïve children with ADHD + ODD and ADHD + CD, and their neural correlates.	Large sample of medication-naïve children with ADHD + ODD, ADHD + CD, and healthy controls; neurocognitive EF tests and functional brain network analyses; correlations between CD symptoms and EF; mediation analysis of SMN(L)–DMN(R) functional connectivity, focusing on younger children.	ADHD + CD patients did not show EF impairments compared with healthy controls; more severe CD symptoms were associated with better EF; in younger children, SMN(L)–DMN(R) connectivity mediated the relation between CD symptoms and inhibition; results suggest ADHD + CD and ADHD + ODD differ in EF profiles, and CD comorbidity may not worsen—and may even improve—EF performance.	Limitations not reported in abstract.
[58]	Biomedicines	To quantify baseline pro- and anti-inflammatory cytokines in prepubertal children with ADHD, examine changes after methylphenidate, and explore interactions with ADHD presentation and ODCD/anxiety comorbidities.	31 prepubertal children with ADHD (13 inattentive, 18 combined); screened for oppositional defiant conduct disorder (ODCD) and anxiety; blood samples at 09:00 before and after ~4.6 months of MPH; IL-1β, IL-5, IL-6, TNF-α, IL-4, IL-10, IL-13 measured via Luminex; factorial analyses.	Overall, MPH did not induce significant cytokine changes; when stratified by ADHD presentation, differences appeared mainly in ADHD-inattentive, often opposite to ADHD-combined; interactions with comorbid factors emerged, with IL-1β and IL-13 showing strongest effects; cytokine differences were probably related to ODCD and diminished after treatment.	Limitations not reported in abstract.

Table 2 (continued)

Authors (Year)	Journal	Aims	Methodology	Findings	Limitations
[65]	Frontiers in Psychiatry	To examine whether reward-related inhibitory control, approach motivation, and autonomic reactivity to reward cues predict later ADHD while accounting for ODD and CU symptoms.	198 preschoolers (mean age ~ 58 months) followed up at ~ 8 years; ADHD diagnosis by clinical interview; ODD symptoms and CU traits via multi-informant ratings; RRIC assessed with Snack-Delay and Gift-Bag tasks; approach motivation tasks; autonomic reactivity via electrodermal activity; longitudinal analyses.	Low reward-related inhibitory control and low autonomic reactivity were uniquely linked to developing ADHD; associations between approach motivation and ADHD were fully explained by comorbid ODD and CU symptoms; high approach motivation marked ADHD with ODD/CU problems, whereas low RRIC and low autonomic reactivity characterized "pure" ADHD, supporting neurocognitive subtypes among externalizing disorders.	Limitations not reported in abstract.
[71]	Psychoneuroendocrinology	To test whether ADHD symptoms relate to diurnal cortisol output when accounting for comorbid symptoms, and to explore effects of substituting ODD for CD in models.	138 adolescents (11–15 years); dimensional measures of ADHD, CD, anxiety, depression (and in sensitivity analyses, ODD instead of CD); saliva samples morning, afternoon, evening across two days; cortisol indices: CAR, AUCg, timepoint levels; regression models adjusting for comorbid symptoms.	After controlling for CD, anxiety, and depression, ADHD symptoms were associated with higher cumulative diurnal cortisol (AUCg), higher morning, and higher afternoon cortisol; CD, anxiety, and depression symptoms were not significantly related to cortisol metrics; in sensitivity analyses, ODD symptoms were associated with lower AUCg and lower morning cortisol, indicating distinct HPA-axis profiles for ADHD vs. externalizing symptoms.	Limitations not reported in abstract.
[12]	Human Brain Mapping	To compare brain morphology (subcortical and cortical) in boys with ADHD-only vs. ADHD + ODD/CD and typically developing peers.	Structural MRI study of 36 ADHD-only boys, 26 ADHD + ODD/CD boys, and 30 typically developing boys aged 11–17, matched for age, pubertal status, and IQ; FreeSurfer analyses of subcortical and cortical volume, cortical thickness, and surface area.	Both ADHD groups showed reduced total gray matter volume and total cortical surface area compared with controls; ADHD + ODD/CD had thicker cortex in a right rostral middle frontal region, and this thickness was positively related to ODD/CD symptom severity even when controlling for ADHD symptoms; no group differences emerged in local cortical volume or surface area; findings suggest delayed cortical thinning and greater impairment in comorbid ADHD + ODD/CD.	Limitations not reported in abstract.
[60]	International Journal of Developmental Neuroscience	To evaluate hot (emotional) and cold (cognitive) executive function in medication-naïve children with ADHD with and without ODD compared to typically developing peers.	Cross-sectional study of 45 children aged 7–12: ADHD + ODD ($n = 15$), ADHD-only ($n = 15$), typically developing ($n = 15$); CSI-4 for behavioral/emotional symptoms; cold EF assessed with WAIS-R Digit Span, Corsi Block, WCST; hot EF with Delay Discounting Task and Iowa Gambling Task.	Relative to typically developing children, ADHD groups showed significantly lower total and backward Digit Span and Corsi scores (cold EF) and lower DDT and IGT scores (hot EF); comorbid ODD did not significantly affect hot or cold EF beyond ADHD; authors conclude both hot and cold EF are impaired in ADHD and recommend cognitive rehabilitation interventions.	Limitations not reported in abstract.

bilateral amygdala volume was correlated with ADHD at-risk status and with inattention and hyperactivity/impulsivity, independent of comorbidities and motion (notably, the symptom association did not extend beyond oppositional-defiant symptoms). This suggests that amygdala volume plays a role in assessing ADHD risk and severity in this demographic [70]. Task-based fMRI in a substantial community cohort supported a bifactor model, revealing that diminished right amygdala activity in response to fearful faces correlated with general externalizing issues, whereas ADHD scores were associated with increased right nucleus accumbens activation to both fearful and happy faces, indicating shared and ADHD-specific limbic–striatal correlates [68].

In a study of 48 clinically referred children aged 5–12 with ADHD and conduct problems, the reward-positivity ERP approximately 300 ms post-feedback was influenced by a Conduct Disorder × irritability interaction, showing increased reward positivity when irritability was high and CD symptoms were low; callous-unemotional traits did not affect these associations [67]. A translational perspective synthesized converging amygdala and corticostriatal alterations in ADHD and ODD/CD, along with evidence that methylphenidate can enhance task-related activation in inhibitory/time-discrimination networks, thereby contextualizing limbic findings within a broader externalizing framework [73]. A complexity analysis in preadolescents revealed diminished sample entropy (resting-state fMRI) in overlapping nodes of cool/hot executive networks, specifically the bilateral superior frontal gyrus, anterior/posterior cingulate gyrus, and bilateral caudate, in cases of comorbidity-free ADHD and comorbidity-free ODD. This was accompanied by more pronounced reductions in ADHD and an exacerbation of deficits when ADHD coexisted with other psychiatric diagnoses [69]. Collectively, these studies—encompassing connectivity, morphometry, task reactivity, and complexity—demonstrate that oppositionality in ADHD is rooted in limbic–striatal circuitry: ADHD-associated amygdala–striatal hypercoupling and caudate-linked network simplification are evident, with the incorporation of oppositional traits corresponding to modifications in these patterns rather than a consistent intensification, and linked to irritability-dependent reward mechanisms.

Theme 2: cerebellar and large-scale network alterations in ADHD + ODD

Three studies indicate cerebellar involvement and extensive network modifications associated with oppositionality in ADHD [13, 55, 69]. In an imaging study involving 27 individuals with ADHD only, 27 with ADHD and ODD, and 27 healthy controls (ages 9–14 years), it was found that those with ADHD and ODD exhibited reduced gray matter volume in the anterior cerebellar regions (Cerebellum_4_5). Additionally, there was increased resting-state functional connectivity from these cerebellar nodes to various cortical areas, including hubs of executive function and the default mode network. The authors suggested that these alterations may worsen the emotional and cognitive deficits linked to ADHD in the presence of ODD [13]. A preadolescent resting-state complexity study revealed diminished voxel-wise sample entropy within both cool and hot executive-function networks for individuals with comorbidity-free ADHD and ODD in overlapping regions (bilateral superior frontal gyrus, anterior/posterior cingulate gyrus, bilateral caudate). ADHD exhibited additional effects in the right inferior/middle frontal gyrus and bilateral frontal orbital cortex. When ADHD co-occurred with other psychiatric diagnoses, entropy reductions extended beyond the comorbidity-free ADHD pattern, suggesting that comorbidity enhances network simplification [69]. In addition to these findings, amygdala-seeded analyses in a large clinical sample showed that ADHD-only (not ADHD + ODD) had more functional coupling from amygdala subregions to the frontal cortex and striatum and higher streamline density in major interhemispheric and projection pathways. These group differences, which show widespread fronto-striatal–limbic coupling in ADHD-only, suggest that adding ODD is linked to a different network topology rather than a simple linear increase [55]. In these studies, the consistent findings include a reduction in regional cerebellar volume in ADHD + ODD compared to ADHD-only, accompanied by cerebello-cortical hyperconnectivity [13], widespread decreases in spontaneous signal complexity within executive networks with comorbidity-related amplification [69], and modified coupling patterns between the fronto-striatal and amygdala when ODD is either present or absent [55].

Theme 3: executive-function phenotypesvhot, cold, and timing

Five studies indicate a prevalence of broader executive dysfunction when oppositional defiant disorder (ODD) coexists with attention-deficit/hyperactivity disorder (ADHD), with variations depending on task domain and sample characteristics [59, 60, 62, 73, 76]. In a study of adolescents (mean age 16 years; $n = 82$ ADHD-only, $n = 82$ ADHD + ODD, $n = 82$ controls) matched for age, gender, IQ, and ADHD type, those with ADHD + ODD exhibited abnormalities in inhibition, working memory, facial emotion recognition, and temporal processing. In contrast, the ADHD-only group demonstrated impairments restricted to working memory and time production.

This suggests that ODD contributes significantly to the executive-function deficits observed in ADHD [59]. In a clinical cohort of 8–12-year-olds classified as ADHD ($n = 246$), ADHD + ODD ($n = 212$), ADHD + CDS ($n = 176$), and controls ($n = 207$), the ADHD + ODD group demonstrated inferior performance in the Central Nervous System Vital Signs (CNSVS) domain compared to other groups, with the exception of psychomotor speed. Their challenges encompassed family communication/problem-solving and the recognition/understanding of others' emotional responses, while the ADHD + CDS group exhibited the lowest psychomotor speed and diminished reaction time/cognitive flexibility in comparison to ADHD [72]. In a smaller, medication-naïve cohort ($n = 45$; ages 7–12), both hot and cold executive functions were compromised in individuals with ADHD compared to typically developing peers; however, comorbid Oppositional Defiant Disorder (ODD) did not further diminish Wechsler digit span, Corsi block, or Iowa Gambling/Delay tasks. Disregarding performance in comparison to ADHD-only [60], indicates that sample size, medication status, and task selection may affect the identification of ODD-related executive function deficits. A clinical study comparing diagnostic groups indicated that children with disruptive mood dysregulation disorder exhibited more severe emotion regulation difficulties in daily life than those with ADHD, demonstrated clinically elevated cognitive flexibility scores relative to normative data, and experienced fewer working memory issues compared to ADHD. No significant differences in daily life executive function were observed between DMDD and ODD, and performance-based executive function tasks showed no variation by diagnosis. Additionally, increased irritability was associated with heightened emotional dysregulation and cognitive inflexibility [63]. In medication-naïve children categorized into ADHD with ODD and ADHD with CD, no executive function (EF) impairments were detected in the CD group compared to controls. Moreover, increased CD severity was associated with enhanced EF. A mediation analysis indicated that the relationship between CD symptoms and inhibition was mediated by left sensorimotor–right default-mode functional connectivity in younger children, suggesting that comorbid disruptive phenotypes may uniquely influence EF networks [76]. Collectively, these studies indicate a consistent broader executive function (EF) impairment associated with oppositional defiant disorder (ODD) in conjunction with attention-deficit/hyperactivity disorder (ADHD) at the group level within larger adolescent and school-age populations [59, 72], variability observed in smaller medication-naïve cohorts [60], and supplementary evidence indicating that irritability and disruptive symptomatology influence the emotional regulation and cognitive flexibility dimensions of EF [63, 76].

Theme 4: neuroendocrine and autonomic markers HPA activity and sympathetic reactivity

Three studies converge on the significance of HPA-axis activity and sympathetic reactivity, with patterns dependent on sex, presentation, and comorbidity [19, 65, 71]. In a medication-naïve cohort (98 ADHD; 107 controls), hair cortisol concentration served as an indicator of long-term HPA activity, while skin-conductance responses reflected sympathetic reactivity. Boys with predominantly inattentive ADHD exhibited lower hair cortisol levels compared to healthy boys, girls with combined-type ADHD demonstrated elevated hair cortisol levels relative to healthy girls, and boys with ADHD plus ODD/CD displayed diminished sympathetic reactivity. When we took into consideration adverse events that happened in childhood, the differences in hair cortisol levels were not significant, but the differences in sympathetic reactivity stayed the same [19]. In a study of community adolescents ($N = 138$; ages 11–15) conducted over two days, ADHD symptoms were linked to elevated cumulative diurnal cortisol (AUCg), as well as increased morning and afternoon cortisol levels, after controlling for symptoms of conduct disorder, anxiety, and depression. Sensitivity analyses substituting conduct disorder with ODD revealed that ODD symptoms were correlated with reduced AUCg and diminished morning cortisol, underscoring the distinct endocrine profiles associated with ADHD and oppositional symptoms [71]. A longitudinal study tracking preschoolers until age eight found that low reward-related inhibitory control and low autonomic reactivity (measured by electrodermal activity) were specifically linked to the development of ADHD. In contrast, the relationship between approach motivation and ADHD was entirely accounted for by oppositional defiant disorder (ODD) and callous-unemotional symptoms, positioning autonomic indices within the developmental trajectories of oppositionality [65]. Collectively, these studies demonstrate that sympathetic blunting is a characteristic of boys with ADHD and comorbid ODD/CD [19], that ODD symptoms correlate with reduced morning and cumulative cortisol levels in community samples [71], and that autonomic markers distinguish ADHD pathways in the presence of oppositional and callous-unemotional traits [65].

Theme 5: immune–metabolic pathways tryptophan kynurenine and cytokines

Two studies directly investigate peripheral immune–metabolic pathways in ADHD with disruptive comorbidity [57, 58]. In a medication-free cohort (46 ADHD-C; 43 ADHD-C + ODD; 33 ADHD-C + CD; 50 controls), serum kynurenine was elevated in ADHD-C compared to controls, while 3-hydroxykynurenine was diminished in ADHD-C and ADHD-C + ODD relative to controls.

The kynurenic acid/kynurenine ratio was increased in ADHD-C, ADHD-C + ODD, and ADHD-C + CD, and the 3-hydroxykynurenine/kynurenine ratio was decreased in ADHD-C, indicating a shift in tryptophan metabolism that did not distinguish ODD from ADHD-C across all metrics [57]. In prepubertal inattentive ADHD treated with methylphenidate for a mean of 4.63 ± 1.87 months (time-of-day sampling at 09:00), interleukin changes were predominantly non-significant as a whole, but when grouped by presentation, differences existed primarily in those with ADHD-inattentive frequently in contrast to ADHD-combined and interacted with comorbid factors; IL-1β (p = .01) and IL-13 (p = .006) reached the highest degree of statistical significance, and differences were associated with the existence of oppositional-defiant-conduct comorbidity and diminished after treatment [58]. These findings indicate that tryptophan kynurenine deviations are associated with ADHD across disruptive subgroups, while presentation-specific cytokine profiles may be influenced by oppositional defiant/conduct disorder comorbidity and are alleviated following stimulant treatment.

Theme 6: dyadic neural synchrony and family/temperament moderators

Three studies concentrate on socio-affective moderators at neural and familial levels [62, 72, 75]. Utilizing hyperscanning fNIRS in 72 mother-child dyads (35 with ODD; 37 healthy), interpersonal neural synchrony exhibited variations between free play and positive/negative discussions in frontal, parietal, and temporal regions, being significantly diminished in ODD dyads. This synchrony was inversely correlated with ODD symptoms and associated with distinct nodes in the ODD symptom network through graph-analytic methodologies [75]. In a cross-sectional clinical sample, maternal anxious/irritable temperament, child inattention/hyperactivity–impulsivity, and child obsessive–compulsive traits exhibited positive correlations with ODD scores, while parental acceptance/involvement demonstrated a negative correlation with ODD. For conduct disorder scores, maternal irritable and paternal cyclothymic/hyperthymic temperaments, along with child ODD and autistic traits, were identified as positive correlates, whereas maternal ADHD was recognized as a negative correlate, highlighting the significance of parental affect and style [62]. In school-age children, family communication and problem-solving difficulties were more pronounced in those with ADHD and cognitive disengagement syndrome, as well as in those with both ADHD and ODD, compared to control groups. Additionally, challenges in recognizing and comprehending others' emotional responses were observed, highlighting the interplay between family dynamics and emotional understanding in conjunction with executive findings [72]. In these three studies, diminished parent–child neural synchrony, particular parental temperament profiles, and family-system functioning correlate with the severity and topology of oppositional symptoms.

Theme 7: treatment responsiveness as a biological readout

Three studies establish a correlation between biological markers and treatment outcomes [58, 66, 73]. In a naturalistic single-dose methylphenidate challenge involving drug-naïve youth (N = 53; 26 ADHD + ODD; 27 ADHD without ODD), reaction-time indices of attention exhibited greater improvement in the ADHD without ODD group compared to the ADHD + ODD group. The groups were comparable at baseline in terms of attention, demographics, and symptom severity, yet they differed in the Dysregulation Profile and disruptive behaviors, which the authors suggested may affect responsiveness to methylphenidate [66]. After approximately 4.6 months of methylphenidate treatment for prepubertal inattentive ADHD, cytokine differences associated with oppositional-defiant-conduct comorbidity (specifically IL-1β and IL-13) were no longer present, suggesting the alleviation of immune deviations through stimulant therapy [58]. A translational synthesis highlighted evidence indicating that methylphenidate augments activation in the bilateral inferior frontal cortex/insula during inhibition and temporal discrimination in pediatric ADHD, while also addressing broader circuit normalization [73]. Within the constraints of these studies, stimulant-associated enhancements in attention are diminished by oppositional comorbidity, whereas inflammatory abnormalities related to ODD/CD seem to normalize during stimulant therapy.

Theme 8: symptom networks and developmental bridges

Four studies concentrate upon symptom-level architectures pertinent to oppositionality in ADHD [8, 61, 64, 65]. A cross-lagged panel network study involving Chinese children aged 6–13 years (N = 263) with ODD and/or ADHD, assessed one year apart, revealed a stable network structure. It identified "annoy" and "blame" as bridging symptoms connecting ODD and ADHD; "annoy" reciprocally predicted "interrupts/intrudes," "blame" predicted "close attention," and "vindictive" predicted various ADHD symptoms, while "angry" was predicted by multiple ADHD symptoms, indicating bidirectional activation across networks [64]. In a substantial clinical cohort (N = 2,015 ADHD; 745 controls), emotional lability scores were elevated in individuals with ADHD compared to controls, regardless of gender or IQ. These scores were highest in those with ADHD and Oppositional Defiant Disorder (ODD) as well as in those with a combined presentation of ADHD. The scores demonstrated significant screening efficacy in Receiver

Operating Characteristic (ROC) analyses, exhibited a declining trend with age in controls, but a persistent trend in ADHD. Additionally, ODD symptoms displayed an increasing trend with age [8]. Among school children aged 6–15 years ($N = 216$), ADHD was correlated with significantly heightened symptoms of ODD, conduct disorder, and anger in comparison to controls, with externalizing behaviors being more pronounced in boys; age did not influence the outcomes [61]. In a longitudinal context, preschool assessments demonstrated that approach motivation associated with subsequent ADHD was entirely elucidated by ODD and callous-unemotional symptoms, suggesting the involvement of oppositional characteristics in developmental trajectories [65]. Collectively, these studies demonstrate that bridge nodes ("annoy," "blame," "vindictive," "angry") and developmental trajectories in emotional lability and oppositional symptoms interweave ADHD and ODD symptoms into a mutually reinforcing network over time.

Theme 9: heterogeneity and comorbidity architectures

Six studies employing various methodologies delineate the heterogeneity and structural positioning of ODD within ADHD [12, 56, 68, 69, 73, 76]. A bifactor model delineating general externalizing traits from specific CD, ODD, ADHD, and callous-unemotional traits exhibited optimal fit in a community sample of 9-year-olds ($N = 11{,}878$). This model indicated a diminished right amygdala response to fear as the general factor and a correlation between ADHD scores and heightened right nucleus accumbens activation [68]. Adolescent boys with ADHD-only and ADHD + ODD/CD exhibited diminished total gray matter and total surface area. In contrast, ADHD + ODD/CD demonstrated a thicker right rostral middle frontal cortex associated with more pronounced ODD/CD symptoms; however, no significant differences were observed in local cortical volume/surface area [12]. In resting-state fNIRS, medication-naïve ADHD with ODD did not exhibit significant differences from ADHD without comorbidity regarding ALFF in prefrontal and temporal regions. Conversely, ADHD with learning disabilities demonstrated elevated ALFF in medial prefrontal, temporal, and left ventrolateral prefrontal channels. This indicates that ODD-related symptoms may be integrated within ADHD rather than representing a distinct fNIRS-defined pathophysiology in that dataset [56]. In medication-naïve children, executive function (EF) profiles exhibited differences between ADHD with CD) and ADHD with ODD. Furthermore, in younger children, the relationship between CD symptoms and inhibition was mediated by the connectivity between the sensorimotor network (SMN(L)) and the default mode network (DMN(R)), suggesting network-level determinants of symptom-specific EF profiles [76]. Resting-state complexity analyses revealed overlapping entropy reductions in comorbidity-free ADHD and ODD. In contrast, broader reductions and extensions across regions of interest (ROIs) were observed when ADHD co-occurred with other diagnoses, suggesting that comorbidity exacerbates dyscomplexity [69]. A translational synthesis highlighted overlapping yet distinct cortico-limbic–striatal substrates associated with ADHD and ODD/CD, alongside stimulant-sensitive networks in ADHD [73]. In summary, these studies characterize ADHD + ODD as a unique configuration within a broader externalizing framework: certain imaging contrasts reveal minimal separation [56], while others indicate region-specific thickness variations associated with ODD/CD symptomatology [12]. Furthermore, extensive modeling attributes oppositional characteristics to both common externalizing variance and symptom-specific patterns [68, 69].

Discussion

The results of this systematic literature review (SLR) provide significant insights into the neurobiological, neurocognitive, and endocrine characteristics of ADHD, both with and without comorbid ODD. The central assertion of this systematic literature review is that oppositionality in ADHD is rooted in limbic–striatal circuitry, complemented by distinct cerebello-cortical and executive overlays. This aligns with and refines conclusions from previous global reviews. Extensive meta-analytic and mega-analytic evidence has associated fronto-striatal–limbic systems with ADHD, characterized by diminished subcortical volumes (amygdala, accumbens, caudate, putamen) and modified task-evoked responsivity [18, 44, 77, 78]. This synthesis enhances the existing literature by comparing ADHD-only with ADHD + ODD. The fact that ADHD-only has stronger amygdala–striatal–frontal coupling than ADHD + ODD and higher streamline density in projection and callosal pathways goes against a simple "dose–response" externalizing account and supports topology shifts with comorbidity. This aligns with ODD/CD-focused imaging meta-analyses indicating amygdala-centric abnormalities that are not congruent with ADHD [1], and with bifactor studies attributing reduced right amygdala fear reactivity to general externalizing while associating ADHD with ventral-striatal activation [68, 77]. Recent normative "brain-chart" analyses that identify amygdala morphometry as an ADHD-risk correlate contextualize the present findings within population-based trajectories rather than clinical convenience samples [70]. The pattern aligns with hierarchical dimensional frameworks (HiTOP/p-factor), wherein ODD variance is associated with antagonistic/disinhibited spectra that overlap, yet are not identical to, ADHD [79–81].

Cerebellar contributions in ADHD + ODD, particularly diminished anterior lobule volume coupled with

hyperconnectivity to executive/default hubs, align with early structural meta-analyses that identify cerebellar reductions as a significant ADHD marker [78], and longitudinal studies that associate a smaller cerebellum with negative outcomes [82], while introducing the comorbidity-specific phenomenon of cerebello-cortical "over-wiring" in ADHD + ODD. Simultaneous findings of diminished resting-state signal complexity within cool/hot executive function networks correspond with systems-level analyses indicating default-mode dysregulation and extensive network instability in ADHD [44, 83]. The current review posits that ODD does not merely exacerbate ADHD-related dysconnectivity but rather reconfigures it. This perspective effectively reinterprets previous null contrasts in prefrontal hemodynamics between ADHD with and without ODD on fNIRS as indicative of potential measurement-modality limitations rather than an actual absence of effect [56].

The executive-function theme combines different primary studies with two consistent signals from larger SLRs: medium-sized, general EF deficits in ADHD that get better with age [84, 85], and disorder-specific impairments in ODD/CD across "hot" EF (decision-making, emotion recognition) [86]. The current systematic literature review (SLR) indicates a more pronounced executive function (EF) impairment in adolescents with both ADHD and ODD compared to those with ADHD alone, aligning with previous meta-analyses. In contrast, small, medication-naïve cohorts exhibiting minimal ODD increments underscore prevalent threats to precision, including underpowered tasks, construct impurity, and task impurity confounds. Parallel emotion-processing syntheses indicate that emotion-recognition deficits in ADHD are identifiable yet heterogeneous, likely influenced by task demands and bottom-up response requirements [84], thereby reinforcing this review's focus on irritability as a transdiagnostic amplifier of EF-adjacent dysfunction.

Endocrine and autonomic findings suggest that lower morning and cumulative cortisol levels associated with ODD-symptoms, alongside diminished sympathetic reactivity in boys with ADHD and comorbid ODD or CD, align with a broader framework of reduced electrodermal and hypothalamic-pituitary-adrenal (HPA) responses in disruptive behavior [87, 88]. These findings also update ADHD-specific meta-analyses that indicate minor and inconsistent cortisol variations when comorbidity is not differentiated [26]. Recent systematic literature reviews (SLRs) converge on hypothalamic-pituitary-adrenal (HPA) axis alterations in pediatric ADHD, yet they caution that effect sizes depend on sampling, assay timing, and adversity covariates [88], the very moderators this SLR addressed by modeling adverse childhood events, resulting in persistent sympathetic effects, but not cortisol effects. This selective robustness bolsters the assertion that autonomic blunting serves as a superior biological marker of oppositionality in ADHD compared to tonic cortisol.

Peripheral immune-metabolic evidence is still not final. The current synthesis accurately categorizes tryptophan-kynurenine shifts as ADHD-broad (rather than ODD-specific) and cytokine variations as sensitive to presentation and comorbidity. This position corresponds with systematic reviews that identify slight, inconsistent inflammatory increases in ADHD [28, 89–91], and caution against heterogeneity among assays and case-mix. It is biologically plausible that methylphenidate normalizes comorbidity-linked IL-1β/IL-13 differences, as evidenced by stimulant-related network normalization observed in fMRI studies [92], however, causality remains uncertain in the absence of randomized biomarker-stratified trials.

The family/temperament and dyadic-synchrony theme is a notable strength, connecting individual neural indices with interpersonal regulation. Decreased parent–child neural synchrony in ODD dyads enhances existing synchrony reviews [93, 94], by linking effects to symptom-network nodes and augmenting SLRs that outline family-system risk pathways for ODD [95]. Alongside EF/irritability findings, the multilevel perspective aligns with developmental network models that suggest family context influences the interplay between emotion-regulation systems and oppositional behaviors over time. Treatment-responsivity as a biological readout is astutely contextualized. The diminished attentional enhancement to methylphenidate in ADHD + ODD aligns with fMRI meta-analysis indicating that stimulants predominantly enhance inferior frontal/insula inhibition and timing networks [92], suggesting that disruptive comorbidity may reflect persistent network dysfunction less responsive to catecholaminergic modulation alone. The immune deviations associated with ODD/CD diminish with stimulants, suggesting a hypothesis and promoting the development of multimodal biomarker panels. However, previous pharmacological systematic literature reviews indicate that neural changes do not consistently correlate linearly with symptom improvement [44], thus, the identified effect modifiers (irritability, CU traits, family processes) should be prospectively stratified.

Lastly, the symptom-network and heterogeneity architectures combined here fit perfectly with modern dimensional nosology. Longitudinal network analysis identifying "annoy," "blame," and "vindictive" as bridge nodes parallels extensive ADHD/ODD network research highlighting the centrality of oppositional and cognitive-disengagement characteristics [96, 97]. Additionally, the current bifactor evidence resonates with the p-factor literature, attributing fear-related amygdala hypo-reactivity to general externalizing behaviors while distinguishing

ADHD-specific striatal responsivity [79, 81]. This duality elucidates the minimal separation observed in certain imaging contrasts (e.g., resting prefrontal ALFF) between ADHD and ADHD + ODD [56], in contrast to cortical thickness or entropy metrics that indicate comorbidity-sensitive deviations.

Methodologically, the SLR is unique in that it (a) separates ADHD-only from ADHD + ODD across modalities; (b) triangulates task fMRI, resting connectivity/complexity, morphometry, ERP, peripheral biomarkers, and dyadic neurobiology; and (c) interprets findings within dimensional models. The remaining limitations, which are also present in previous SLRs, encompass task and pipeline heterogeneity, limited medication-naïve samples, motion/resolution confounds in pediatric imaging, cross-sectional inference for developmental assertions, and potential publication bias in EEG/ERP and cytokine literatures [98–100]. The most defensible translational message is not that ODD "adds more of the identical" but that it remodels ADHD's neural and physiological landscape toward altered limbic–striatal coupling, cerebello-cortical compensation, broader EF and emotion-regulation burdens, and reduced autonomic reactivity within a general externalizing architecture. Future trials should utilize these insights to examine mechanism-guided combinations (stimulants combined with irritability-targeted cognitive-behavioral therapy/parent training) and to assess biomarker-defined subgroups.

As the number of studies included was very heterogeneous, especially in methods used to diagnose ADHD and ODD, sample sizes, and the nature of neurobiological and psychological outcomes measures, a formal quantitative synthesis or subgroup analysis was not possible in the review. Methodological recommendations imply that these types of analyses need fairly similar study designs and standard outcome measures that would generate valid and interpretable findings [51, 101]. As a result, an inductive narrative synthesis was used to support the diversity in methodology and to give a wider interpretation of the neurobiological patterns of ADHD-ODD comorbidity. Though narrative synthesis can be perceived to be inappropriate in cases where evidence is heterogeneous and inappropriate to use in meta-analysis, it is regarded as a suitable method when the evidence is heterogeneous and that cannot be utilized in the context of meta-analysis and can be used to build a basis upon which future studies can use more standardized methods and have a similar dataset [101, 102].

Despite significant findings, the present study is limited to include studies that are mostly based on cross-sectional research design. Although these studies can have good information in terms of neurobiological correlates, it limits the possibility of establishing causal relationships as well as the manner in which neural changes occur in various stages of development. Since the progression of ADHD and ODD has dynamic developmental dynamics, future studies ought to focus on large-scale longitudinal studies to monitor neurobiological indicators, as well as individual changes in behaviors and cognition. These longitudinal designs would aid in clarifying the leads up to the presentation of comorbid symptoms or occurring as an outcome of them thus beefing the causal interpretation and addressing the formulation of more specific and developmentally sensitive intervention plans.

The wide area of the question of the review was carefully chosen to include the complex and multifactorial nature of ADHD in the presence of comorbid ODD. In spite of the sound complaint that the reviewer made about the incorporation of several biological, neurological and neurocognitive domains, current studies indicate that ADHD and ODD comorbidity should be viewed in an integrative framework involving multiple interacting neurobiological mechanisms, as opposed to single working mechanisms. Research has demonstrated that dysfunctions, which overlap both emotional regulation, executive functioning, and reward processing parts of the brain, mostly in the prefrontal cortex, amygdala, and frontostriatal loops, are involved in both ADHD and ODD, indicating the necessity to adopt a cross-domain approach [1, 103]. Moreover, neurocognitive aspects, including impairment of inhibitory control, emotional control, and decision-making have been observed to have common underlying neurobiological mechanisms in children with ADHD and disruptive behavior disorders (Matthys et al., 2013). The synthesis of evidence in these areas allowed the current review to determine deeper patterns and converging processes that can lead to the emergence and continuation of ADHD ODD comorbidity. However, it is admitted that future researches can be facilitated by taking particular domains in biological or neurocognitive areas and using domain-related meta-analytic techniques to shed some more light on these associations. This review presents a thorough overview of the existing evidence base despite the limited scope of its conceptualization and shows the significance of combined methods in comprehending the multifaceted mechanisms of the comorbidity of ADHD and ODD.

Recommendations

This systematic literature review supports a biologically grounded, multidomain framework of oppositionality in ADHD, characterized by limbic–striatal signatures (amygdala–striatal coupling, caudate-linked network simplification), cerebello-cortical modifications, increased executive demands, and selective autonomic/HPA dysregulations. In clinical practice, evaluation should transcend mere symptom enumeration to include dimensional profiles of irritability, fluctuating executive

functions, and familial systemic influences, supplemented—when possible—by scalable biomarkers (e.g., amygdala–striatal reactivity indices, resting-state complexity metrics, and electrodermal responsivity). Treatment planning must consider diminished stimulant response in the context of ODD and should integrate pharmacotherapy with parent training/CBT focused on irritability, emotion recognition training, and family communication/problem-solving modules. The findings advocate for the early identification of at-risk dyads (characterized by diminished parent–child neural synchrony and heightened irritability) and the implementation of stepped-care pathways that incorporate both behavioral and pharmacological interventions. Before deciding on a medication, you should get multimodal baselines (clinical + EF batteries + autonomic readouts) and then check them again after titration. Programs should use measurement-based care dashboards to keep track of things like attention levels, irritability, and family communication. These dashboards should have set levels for when to add more support (for example, parent training or school-based support). At the systems level, funding should be provided for integrated clinics that can conduct neurobehavioral assessments alongside family interventions.

Limitations

The evidence base is limited by the diversity of imaging pipelines and tasks, small medication-naïve samples, insufficient power for sex and developmental stratification, motion artifacts in pediatric neuroimaging, inconsistent assay timing in endocrine studies, and probable publication bias in EEG/ERP and cytokine research. Most studies included in this review are cross-sectional, which limits the ability to determine causal relationships or understand the developmental progression of neurobiological changes associated with comorbid ADHD and ODD. Because both conditions evolve across development, future research should prioritize large-scale longitudinal studies, preferably pre-registered, to track neural changes alongside symptom trajectories over time and to provide stronger evidence regarding causality and long-term outcomes. Clinic-based sampling and not enough low- and middle-income settings make ecological validity worse. The research that comes next needs to be Things that are important include: (1) pre-registered, multi-site longitudinal groups that separate ADHD-only, ADHD + ODD, and disruptive subtypes, with harmonized tasks (inhibition, timing, affective decision-making) and prevalent imaging/physiology pipelines; (2) biomarker-stratified randomized trials testing combined stimulant + irritability-focused behavioral packages, with mechanistic mediators (limbic–striatal connectivity, cerebello-cortical coupling, autonomic indices); (3) normative modeling and network methodologies that connect symptom bridge nodes ("annoy," "blame," "vindictive," "angry") to neural and physiological edges over time; (4) developmental windows and sex-specific moderators; (5) the integration of genetics, adversity, and family-level moderators through causal mediation/instrumental-variable techniques; and (6) high-frequency, real-world phenotyping (EMA/wearables) to assess generalization beyond the laboratory setting. Oppositionality in ADHD is not simply an exacerbation of symptoms; it represents a reconfiguration of neural topology and physiology that impairs executive and socio-emotional regulation and alters treatment responsiveness. Anchoring care to dimensional, multimodal profiles and testing mechanism-guided combinations presents the most credible approach to achieving sustainable, developmentally appropriate enhancements.

Conclusion

This review illustrates that ODD comorbidity significantly alters the neurobiological and functional characteristics of ADHD across neuroimaging, neurocognitive, endocrine, autonomic, and immune–metabolic domains. Oppositionality was consistently associated with modified limbic–striatal signaling, cerebello-cortical compensation, diminished executive-network complexity, and unique patterns in the HPA-axis and sympathetic systems. These convergent findings suggest that ODD+ADHD constitutes a neurobiologically distinct phenotype rather than merely a severity variant of ADHD. Clinically, this underscores the imperative of amalgamating stimulant therapy with emotion-regulation, parent–child relational, and behavioral interventions. Subsequent longitudinal and mechanistic investigations are crucial to elucidating developmental trajectories and enhancing precision-tailored care.

Abbreviations

Abbreviation	Full Form
ADHD	Attention-Deficit/Hyperactivity Disorder
ODD	Oppositional Defiant Disorder
CD	Conduct Disorder
DMDD	Disruptive Mood Dysregulation Disorder
CU (traits)	Callous–Unemotional Traits
LD	Learning Disorder
DSM-5	Diagnostic and Statistical Manual of Mental Disorders, 5th Edition
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PICO	Population, Intervention, Comparison, Outcome
SLR	Systematic Literature Review
MEDLINE	Medical Literature Analysis and Retrieval System Online
EMBASE	Excerpta Medica Database
PsycINFO	Psychological Information Database (by APA)

EEG	Electroencephalography
ERP	Event-Related Potential
fMRI	Functional Magnetic Resonance Imaging
rsfMRI	Resting-State Functional MRI
DTI	Diffusion Tensor Imaging
DWI	Diffusion-Weighted Imaging
VBM	Voxel-Based Morphometry
rsFC	Resting-State Functional Connectivity
fNIRS	Functional Near-Infrared Spectroscopy
fNIRS-hyperscanning	Dual-brain fNIRS Hyperscanning
RewP	Reward Positivity (ERP component)
ENIGMA-ASB	Enhancing NeuroImaging Genetics through Meta-Analysis—Antisocial Behavior Working Group
DMN	Default Mode Network
Abbreviation	Full Form
mPFC	Medial Prefrontal Cortex
TPJ	Temporoparietal Junction
MFG	Middle Frontal Gyrus
SFG / SFGdor / SFGmed	Superior Frontal Gyrus (dorsal / medial)
ACC	Anterior Cingulate Cortex
PCC	Posterior Cingulate Cortex
ANG	Angular Gyrus
DMN–frontoparietal interactions	Connectivity between Default Mode Network & Frontoparietal Network
amygdala–mPFC coupling	Functional coupling between Amygdala and Medial Prefrontal Cortex
Abbreviation	Full Form
EF	Executive Function
Cool EF	Non-emotional/cognitive executive function
Hot EF	Emotionally-driven executive function
IQ	Intelligence Quotient
SES	Socioeconomic Status
WCST	Wisconsin Card Sorting Test
Corsi Block	Corsi Block-Tapping Task
WAIS-R Digit Span	Wechsler Adult Intelligence Scale–Revised Digit Span
Delay Discounting	Impulsive choice paradigm
Iowa Gambling Task (IGT)	Iowa Gambling Task
CNSVS	Central Nervous System Vital Signs (neurocognitive test battery)
Abbreviation	Full Form
HPA axis	Hypothalamic–Pituitary–Adrenal Axis
LC–MS/MS	Liquid Chromatography–Tandem Mass Spectrometry
CAR	Cortisol Awakening Response
AUCg	Area Under the Curve with respect to Ground (overall cortisol output)
Abbreviation	Full Form
SNAP-IV	Swanson, Nolan, and Pelham Rating Scale–IV
PHQ-9	Patient Health Questionnaire-9
SCARED	Screen for Child Anxiety Related Emotional Disorders
CBCL	Child Behavior Checklist
Abbreviation	Full Form
RoB 2	Cochrane Risk of Bias 2 Tool (randomized trials)
ROBINS-I	Risk Of Bias In Non-randomized Studies–of Interventions
JBI	Joanna Briggs Institute (critical appraisal tools)
ANCOVA	Analysis of Covariance
ROC	Receiver Operating Characteristic
CLPN	Cross-Lagged Panel Network
Abbreviation	Full Form
ABCD framework/study	Adolescent Brain Cognitive Development Study
THC:CBD	Tetrahydrocannabinol to Cannabidiol ratio

Acknowledgements
We acknowledge the support of Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2026R255), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Authors' contributions
Shah SM: Writing-original draft, Data collection, and investigation process, analyzed the study data. FSA and GSA: Review.

Funding
The author(s) declared that financial support was received for this work and/or its publication. Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2026R255), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Data availability
Data available upon request from corresponding author.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 16 November 2025 / Accepted: 22 April 2026
Published online: 30 April 2026

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